

2) Aim :- To design, stimulate and verify the magnification functionality of various fundamental combinational logic circuits using Verilog HDL.

3) Theory :-

This experiment covers the design and verification of standard combinational circuits and fundamental concepts of the Verilog language.

Tools Required :-

Model Sim :- This is an HDL simulation tool used for verifying the functionality of digital circuits.

Verilog HDL :- This is the language used to describe the digital circuits and their test benches.

★ 4 bit ripple carry adder

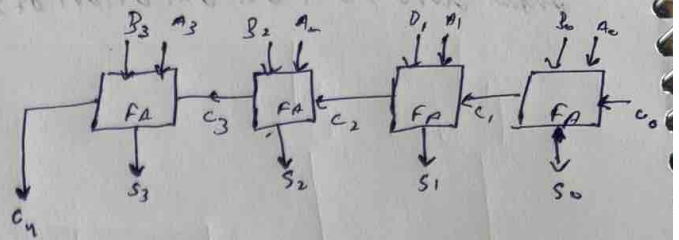
```

module full-adder(S,C,A,B,Cin);
input A,B,Cin;
output S,C;

wire w1,w2,w3;
XOR g0(w1,B,Cin);
XOR g1(S,w1,A);

and g2(w2,w1,A);
and g3(w3,B,Cin);
or g4(C,w2,w3);

endmodule
    
```



```

module ripplecarry-adder(S,C,a,b,Cin);
input [3:0] a,b;
input Cin;
output [3:0] S;
output C;

wire C1,C2,C3;

full-adder m0(S[0],C1,a[0],b[0],Cin);
full-adder m1(S[1],C2,a[1],b[1],C1);
full-adder m2(S[2],C3,a[2],b[2],C2);
full-adder m3(S[3],C,a[3],b[3],C3);

endmodule
    
```

```

module test1-tb();
reg [3:0] a,b;
reg Cin;
wire [3:0] S;
wire C;

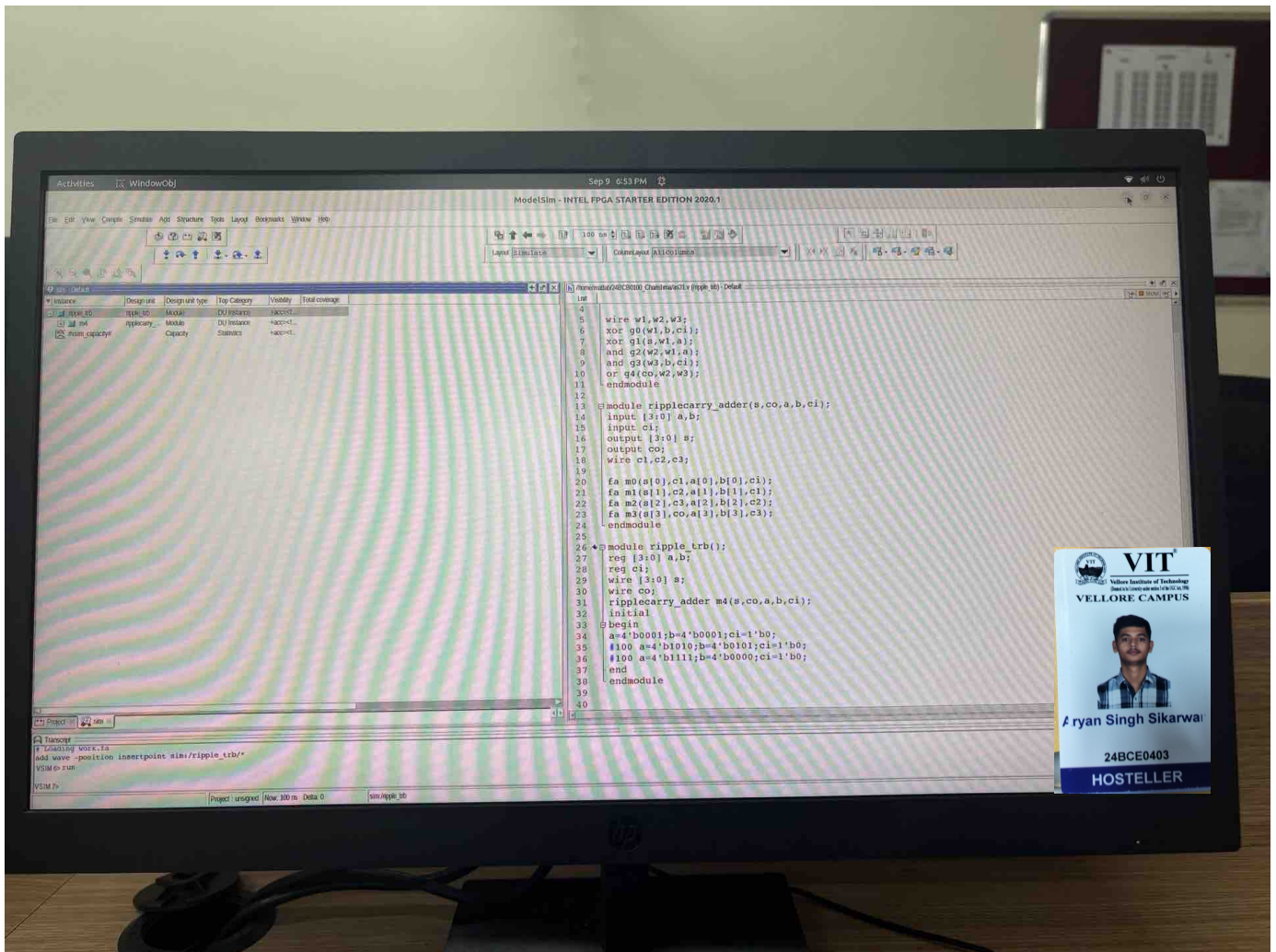
ripplecarry-adder m4(S,C,a,b,Cin);

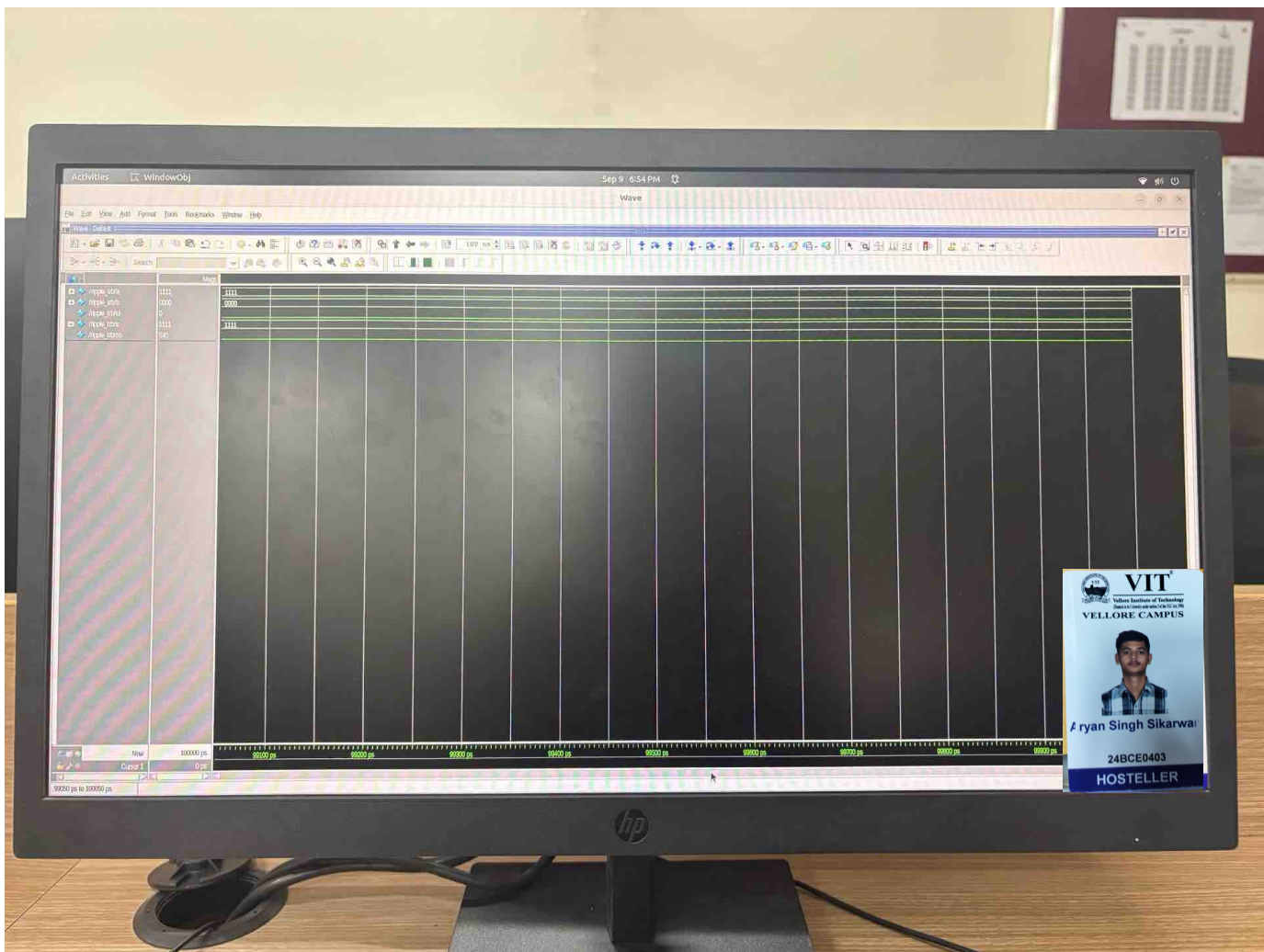
initial
begin
a = 4'b1100; b = 4'b1111; Cin = 1'b0;
# 100 a = 4'b1010; b = 4'b0101; Cin = 1'b0;

end

endmodule
    
```

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★ 3 bit magnitude comparator circuit

```

module com-3bit (E,G,L,A,B);
    input [2:0] A,B;
    output E,G,L;
    wire x0,x1,x2;
    assign x0 = A[0] ^ B[0];
    assign x1 = A[1] ^ B[1];
    assign x2 = A[2] ^ B[2];
    assign E = x2 & x1 & x0;
    assign G = (A[2] & (~B[2])) | (x2 & (A[1] & (~B[1]))) | (x2 & x1 & (A[0] & (~B[0])));
    assign L = (B[2] & (~A[2])) | (x2 & (B[1] & (~A[1]))) | (x2 & x1 & (B[0] & (~A[0])));
endmodule

```

```

module tb-comb();
    reg [2:0] A,B;
    wire E,G,L;
    comb-3bit m0(E,G,L,A,B);
    initial
    begin
        A = 3'b111; B = 3'b111;
        #100 A = 3'b111; B = 3'b001;
        #100 A = 3'b001; B = 3'b111;
    end
endmodule

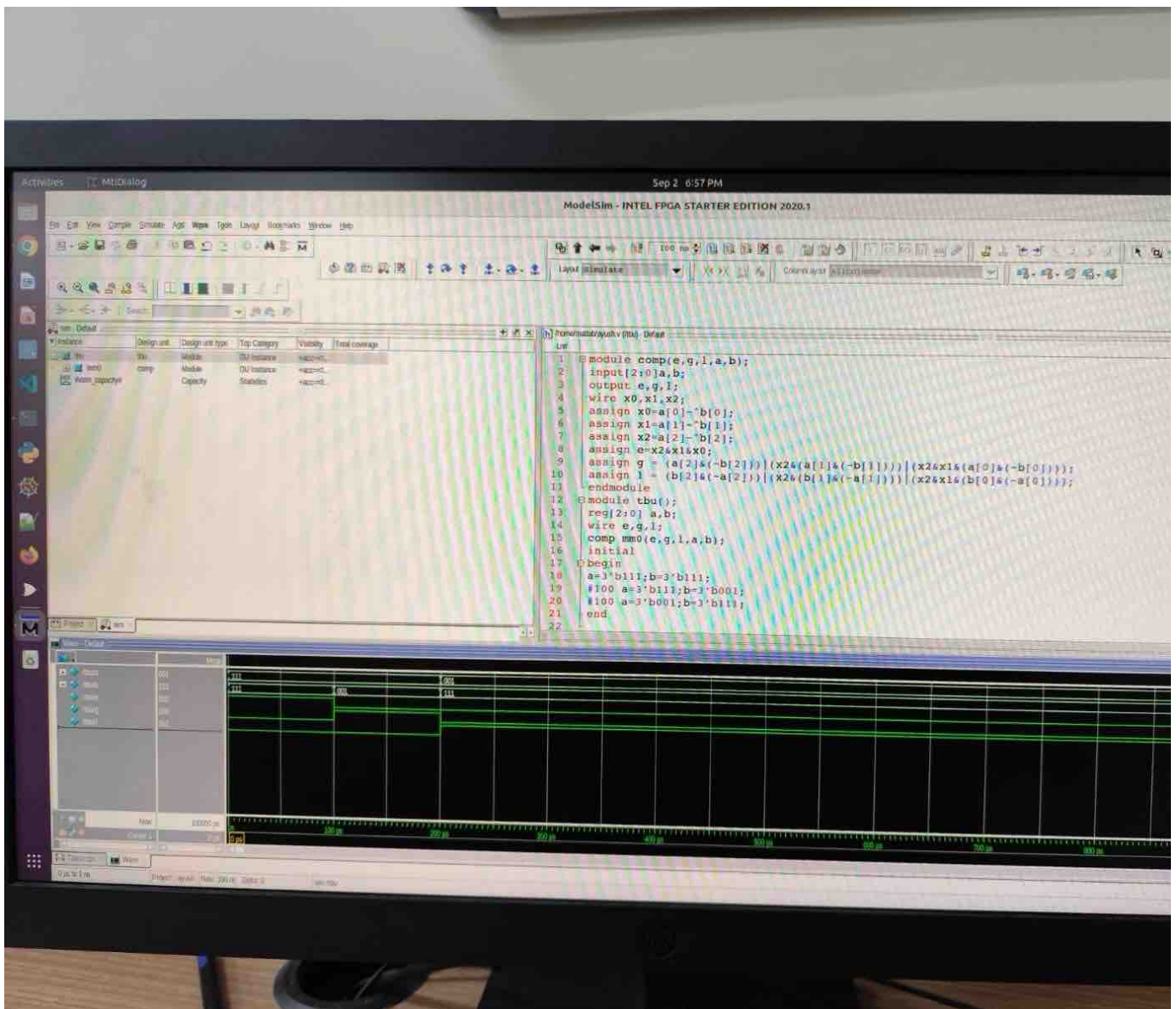
```

A = A₂ A₁ A₀

B = B₂ B₁ B₀

A ₂ A ₁ A ₀	B ₂ B ₁ B ₀	A > B	A = B	A < B
0 0 0	0 0 0	0	1	0
0 0 0	0 0 1	0	0	1
0 0 0	0 1 0	0	0	1
0 0 0	0 1 1	0	0	1
0 0 0	1 0 0	0	0	1
---	---	--	--	--
1 1 1	1 1 0	1	0	0
1 1 1	1 1 1	0	1	0

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★ Two bit by two bit multiplier

```
module HA(S,C,A,B);
```

```
input A,B;
```

```
output S,C;
```

```
XOR m0(S,A,B);
```

```
and m1(C,A,B)
```

```
Endmodule
```

```
module tb();
```

```
reg A,B;
```

```
wire S,C;
```

```
HA m0(S,C,A,B);
```

```
initial
```

```
begin
```

```
A = 1'b0; B = 1'b0;
```

```
# 100 A = 1'b0; B = 1'b1;
```

```
# 100 A = 1'b1; B = 1'b0;
```

```
# 100 A = 1'b1; B = 1'b1;
```

```
End
```

```
endmodule
```

```
include "HA.v" module
```

```
mult-2bit(P,A,B);
```

```
input [1:0] A,B;
```

```
output [3:0] P;
```

```
wire w0,w1,w2,w3;
```

```
and g0(P[0],A[0],B[0]);
```

```
and g1(w0,A[1],B[0]);
```

```
and g2(w1,A[0],B[1]);
```

```
and g3(w3,A[1],B[1]);
```

```
HA m0(P[1],w2,w1,w0);
```

```
HA m1(P[2],P[3],w2,w3);
```

```
Endmodule
```

```
module tb-mult1);
```

```
reg [1:0] A,B;
```

```
wire [3:0] P;
```

```
mult-2bit m0(P,A,B);
```

```
initial
```

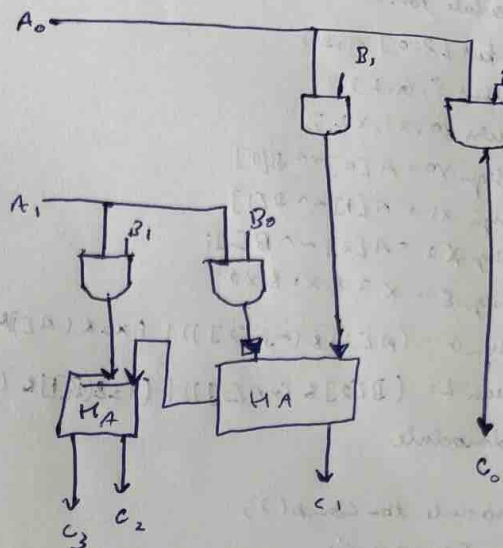
```
begin
```

```
A = 2'b11; B = 2'b11;
```

```
# 100 A = 2'b10; B = 2'b11;
```

```
End
```

```
Endmodule
```



A	B	P
00	00	0000
00	01	0000
00	10	0000
00	11	0000
01	00	0000
01	01	0001
01	10	0010
01	11	0011
10	00	0100
10	01	0101
10	10	0110
10	11	0111
11	00	1100
11	01	1101
11	10	1110
11	11	1111

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Activities Mtdialog

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ModelSim - INTEL FPGA STARTER EDITION 2020.1

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Layout Simulation

Design Library All I/O Objects

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```

1
2 module HA(S,C,A,B);
3   input A,B;
4   output S,C;
5   xor mo(S,A,B);
6   and m1(C,A,B);
7 endmodule
8 module multi_2bit(P,A,B);
9   input[1:0]A,B;
10  output[3:0]P;
11  wire w0,w1,w2,w3;
12  and g0(P[0],A[0],B[0]);
13  and g1(w0,A[1],B[0]);
14  and g2(w1,A[0],B[1]);
15  and g3(w3,A[1],B[1]);
16  HA mm0(P[1],w2,w1,w0);
17  HA mm1(P[2],P[3],w2,w3);
18 endmodule
19 module tb_mult();
20   reg[1:0]A,B;
21   wire[3:0]P;
22   multi_2bit mm3(P,A,B);
23   initial
24   begin
25     A=2'b11; B=2'b11;
26     #100 A=2'b10; B=2'b11;
27   end
28 endmodule

```

Process

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